AAE-E2001 - Computational Fluid Dynamics D

Assignment 1: 1D Convection-Diffusion

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1 Theory Exercise

Equation 1 is the one dimensional convection equation for a scalar c.

$$\frac{\partial C}{\partial x} + \mathrm{u}\frac{\partial C}{\partial x} = 0. \tag{1}$$

The spatial derivative term of $\frac{\partial C}{\partial x}$ is discretized by $\frac{C_i - C_{i-1}}{\Delta x}$ if its coefficient is positive (here the velocity is positive), for the upwind method. In order to demonstrate the order of the discretization, Taylor series has been used as follows:

$$\begin{aligned} C_{i-1} = C(x-\Delta x) &= C(x) - \Delta x\frac{\partial C(x)}{\partial x} + \frac{\Delta x^2}{2!}\frac{\partial^2 C(x)}{\partial x^2} - O(\Delta x^3) \\ \Rightarrow \quad C_{i-1} &= C_i - \Delta x\frac{\partial C_i}{\partial x} + \frac{\Delta x^2}{2!}\frac{\partial^2 C_i}{\partial x^2} - O(\Delta x^3) \\ \Rightarrow \quad \frac{\partial C}{\partial x} &= \frac{C_i - C_{i-1}}{\Delta x} + \underbrace{\frac{\Delta x}{2!}\frac{\partial^2 C_i}{\partial x^2}}_{\text{Leading Error Term}} - O(\Delta x^2) \end{aligned} \tag{2}$$

So from equation 2 it has been shown that the leading error term is of the first order.

The spatial derivative term of $\frac{\partial C}{\partial x}$ is discretized by $\frac{C_{i+1} - C_{i-1}}{2\Delta x}$ for the central difference method. In order to demonstrate the order of the discretization, Taylor series has been used as follows:

$$\begin{aligned} C_{i+1} = C(x+\Delta x) &= C(x) + \Delta x\frac{\partial C(x)}{\partial x} + \frac{\Delta x^2}{2!}\frac{\partial^2 C(x)}{\partial x^2} + \frac{\Delta x^3}{3!}\frac{\partial^3 C(x)}{\partial x^3} + O(\Delta x^4), \\ C_{i-1} = C(x-\Delta x) &= C(x) - \Delta x\frac{\partial C(x)}{\partial x} + \frac{\Delta x^2}{2!}\frac{\partial^2 C(x)}{\partial x^2} - \frac{\Delta x^3}{3!}\frac{\partial^3 C(x)}{\partial x^3} + O(\Delta x^4). \end{aligned} \tag{3}$$

Subtracting the two expansions gives:

$$C_{i+1} - C_{i-1} = 2\Delta x\frac{\partial C}{\partial x} + \frac{2\Delta x^3}{3!}\frac{\partial^3 C}{\partial x^3} + O(\Delta x^5). \tag{4}$$

Rearranging to isolate the first derivative:

$$\Rightarrow \quad \frac{\partial C}{\partial x} = \frac{C_{i+1} - C_{i-1}}{2\Delta x} - \underbrace{\frac{\Delta x^2}{3!}\frac{\partial^3 C}{\partial x^3}}_{\text{Leading Error Term}} + O(\Delta x^4). \tag{5}$$

So from the above equation it is shown that the leading error term of the central difference discretization is of the second order.

The temporal derivative term $\frac{\partial c}{\partial t}$ is discretized using the explicit forward Euler method. Using a Taylor series expansion in time about t^n:

$$
\begin{aligned}
C_i^{n+1} &= C(x, t + \Delta t) \\
&= C(x,t) + \Delta t \frac{\partial C(x,t)}{\partial t} + \frac{\Delta t^2}{2!} \frac{\partial^2 C(x,t)}{\partial t^2} + O(\Delta t^3)
\end{aligned} \tag{6}
$$

Rearranging to isolate the first time derivative gives:

$$
\Rightarrow \quad \frac{\partial C}{\partial t} = \frac{C_i^{n+1} - C_i^n}{\Delta t} - \underbrace{\frac{\Delta t}{2!} \frac{\partial^2 C}{\partial t^2}}_{\text{Leading Error Term}} + O(\Delta t^2) \tag{7}
$$

So from Eq. 7, it is demonstrated the the explicit forward Euler method of the temporal derivative is of first order $O(\Delta t)$.

2 Simulation Exercise

The objective is this section is to model one dimensional (1D) linear advection of a passive scalar T (a dimensionless normalized quantity) in a periodic domain ($0 < x < 2\pi$) by the use of OpenFOAM. Eq. 8 shows the initial condition of this quantity.

$$
T(x, t = 0) = exp(-\frac{(x - x_0)^2}{2\sigma^2}) \tag{8}
$$

In order to perform the simulation cyclic Boundary conditions have been used both at the right and left domain. **functions** solver with **incompressibleFluid** subSolver have been used. A **scalarTransport** function is also used for the passive scalar T. In our simulation the domain size is taken to be $L = 2\pi$, and the velocity is assumed to be 2π m/s. As stated in the **T** file, the value of standard deviation is assumed to be $\sigma = L/15$.

2.1 Part A

In order to determine an appropriate value of the scheme parameter X for different convection-term discretization schemes (blended, MUSCL, and SuperBee), **10?!** simulations were first performed using the SuperBee scheme with X varied from 0 to 1 in uniform increments. It was observed that all simulations produced identical results. The same behavior was also observed for the MUSCL scheme. Eventually, the blended scheme was adopted, for which variations in X led to different results.

To determine the optimal value of X for the **blended** scheme, the same range from 0 to 1 with increments of 0.1 was considered. For these simulations, a total of $N = 40$ computational cells was used. Since it was suggested that a Courant number of approximately 0.5 is sufficient, a time step of $\Delta t = 0.01$ was selected. The corresponding Courant number for this case is:

$$
C_0 = u \frac{\Delta t}{\Delta x} = 2\pi \frac{0.01}{L/40} = 0.4
$$

Figure 1 shows the results obtained with different values of X for the blended scheme after one by-pass time ($T_{\text{simulation}} = L/U = 1$ s). Based on this figure, a value of $X = 0.9$ was selected for subsequent comparisons, as it exhibits the smallest error relative to the exact solution.

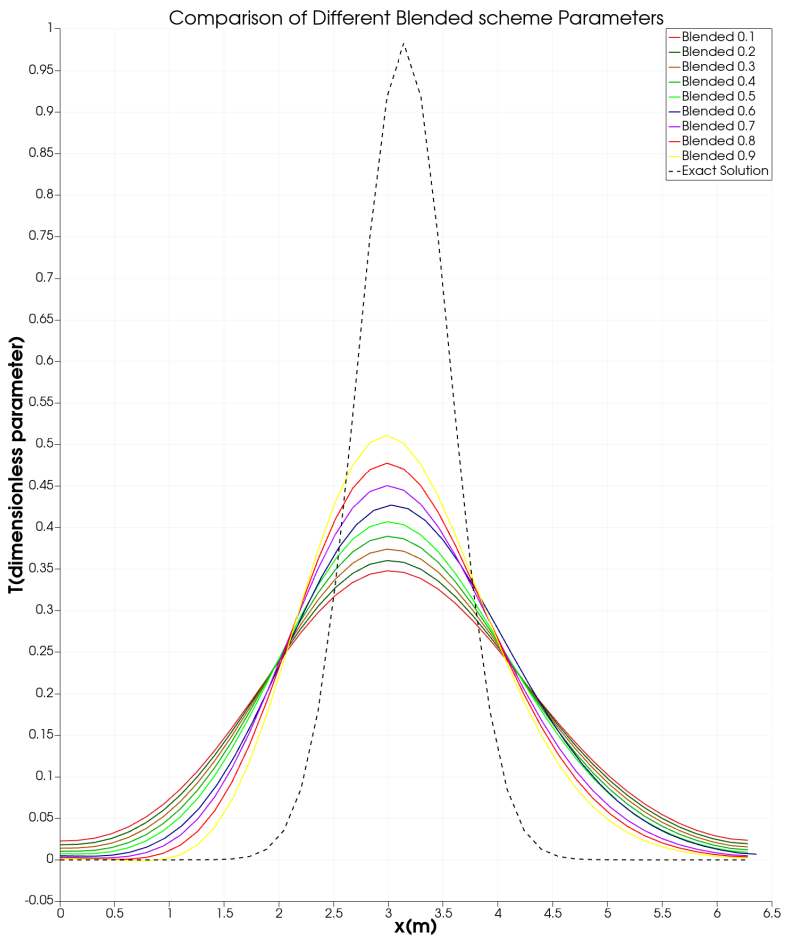

Figure 1: Comparison of results for different values of X in the blended scheme with 40 grid cells, and after one by-pass time

It was later found that the chosen time step ($\Delta t = 0.01$) could lead to stability issues for other discretization schemes when smaller grid sizes were used, as it results in an increased Courant number. To avoid this problem, a reduced time step of $\Delta t = 0.004$ was adopted for all subsequent simulations. With this choice, the Courant number for the smallest grid size considered ($N = 120$) is $C_0 = 0.48$, which would satisfy the stability requirement.

Figure 2 presents a comparison of the passive scalar profiles obtained using different convection discretization schemes on a domain with $N = 40$ cells after one by-pass time. The simulations were performed using the blended scheme with $X = 0.9$, MUSCL with $X = 0.5$, SuperBee with $X = 0.5$, as well as the upwind and linear schemes. A comparison of the results shows that the upwind scheme exhibits strong numerical diffusion, leading to a significant reduction in the peak amplitude and a pronounced spreading of the Gaussian pulse. In contrast, the linear scheme preserves the peak amplitude more effectively than the upwind scheme but introduces noticeable dispersive oscillations near regions with steep gradients.

The SuperBee, MUSCL, and blended schemes belong to the class of total variation diminishing (TVD) schemes and provide a compromise between the dispersive and diffusive errors observed in the upwind and linear schemes. Among these, the SuperBee scheme with $X = 0.5$ demonstrates the best agreement with the analytical solution for this mesh resolution, outperforming both the MUSCL and blended schemes.

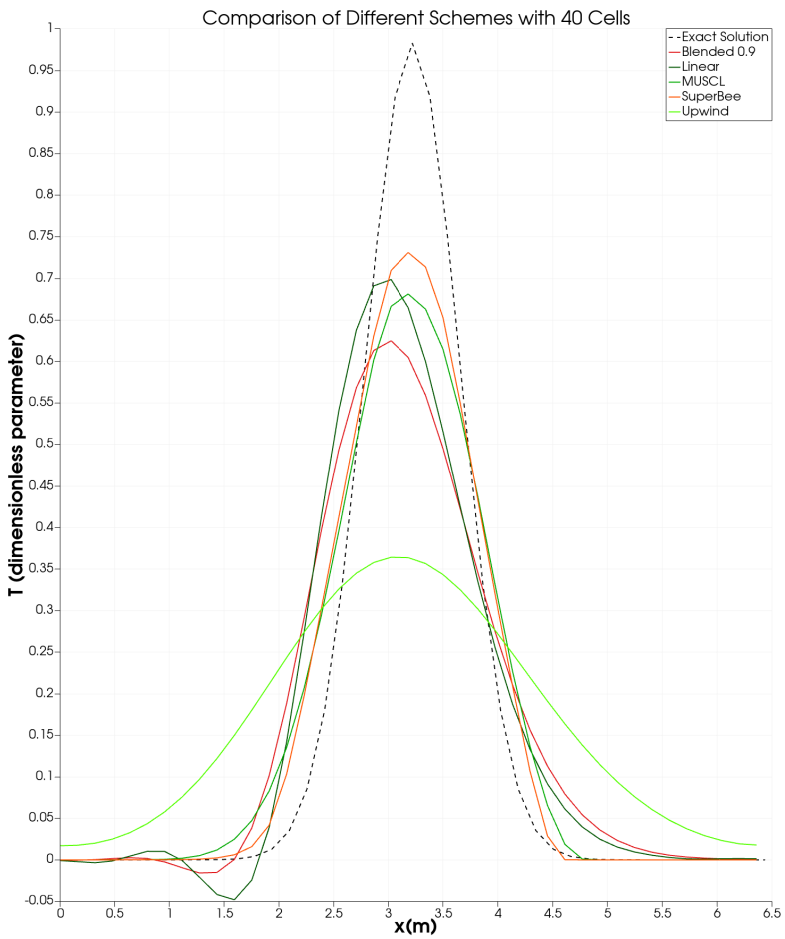

Figure 2: Comparison of results of upwind, linear, blended 0.9, MUSCL 0.5, SuperBee 0.5, and Analytical for 40 cells after one by-pass time

2.2 Part B

Figure 3 shows the results obtained with $N = 80$ cells using the same time step, while Figure 4 presents the results for $N = 120$ cells. As the grid is refined, a clear reduction in numerical error is observed; however, the solution still deviates from the exact solution. This remaining discrepancy is attributed primarily to the chosen time step.

Among the tested schemes, the upwind scheme exhibits the largest error due to its strongly diffusive nature, whereas the SuperBee scheme produces the smallest error in comparison to the analytical solution.

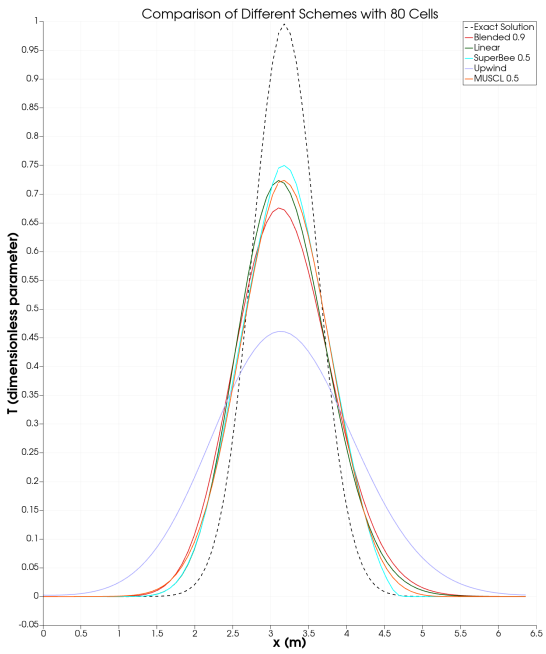

Figure 3: Comparison of results of upwind, linear, blended 0.9, MUSCL 0.5, SuperBee 0.5, and Analytical for 80 cells after one by-pass time

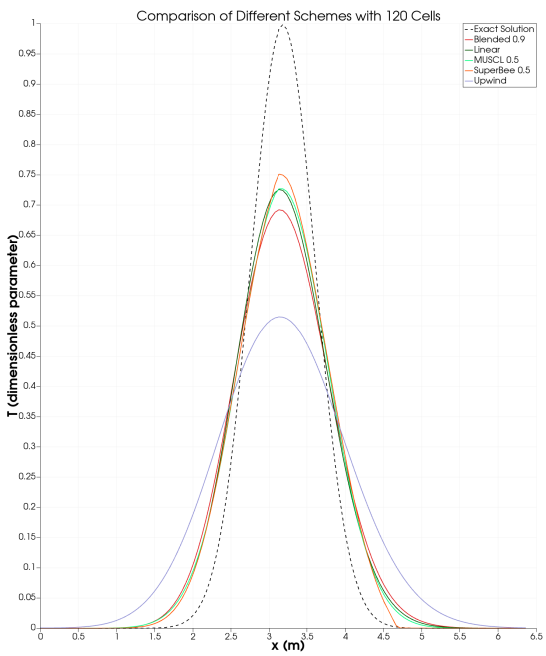

Figure 4: Comparison of results of upwind, linear, blended 0.9, MUSCL 0.5, SuperBee 0.5, and Analytical for 120 cells after one by-pass time

In order to have better results a final simulation has been carried out. In the results shown in Fig 5, it can be seen that The SuperBee TVD scheme shows the best agreement with analytical solution.

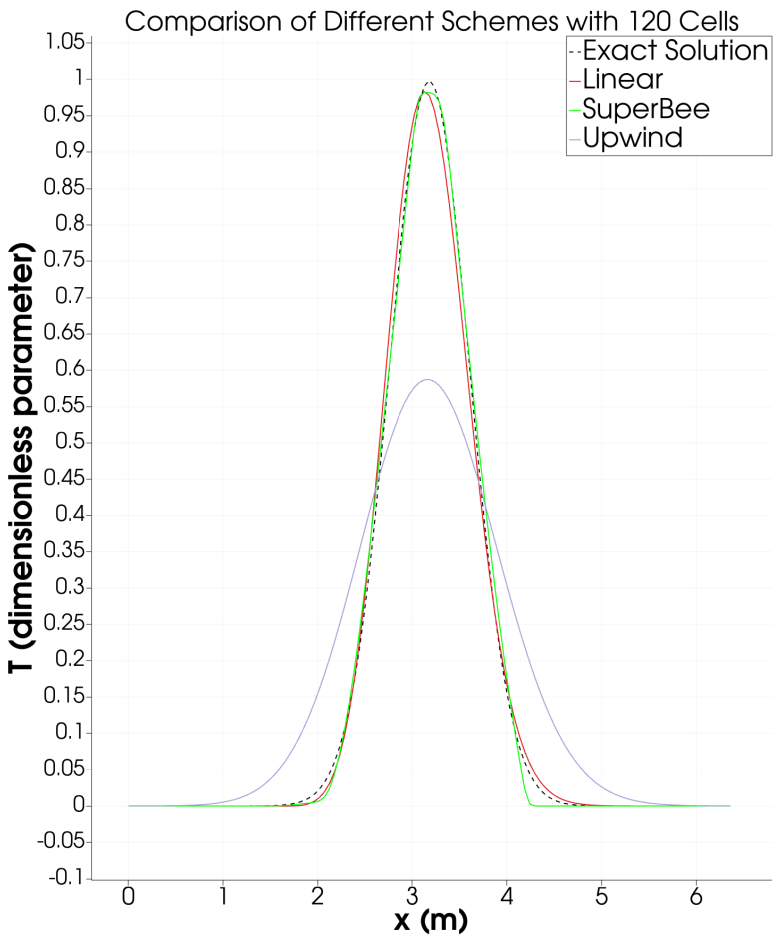

Figure 5: Comparison of results of upwind, linear, and SuperBee 0.5, and Analytical for 120 cells after one by-pass time

2.3 Questions

2.3.1 What is the effect of the variation of the selected scheme parameter? What do you think the parameter values 0.0 and 1.0 represent? Which is the best scheme?

The selected scheme parameter controls the balance between numerical diffusion and higher-order accuracy. Lower values of the parameter result in stronger limiting, making the scheme more diffusive but also more stable, while higher values allow a larger contribution from the higher-order discretization, reducing diffusion but increasing the risk of dispersive oscillations.

A parameter value of 0.0 corresponds to the most diffusive behavior, effectively reverting the scheme toward a first-order upwind discretization, whereas a value of 1.0 represents the least restrictive limiting and a stronger contribution from the higher-order scheme. Based on the obtained results, the SuperBee scheme provides the best overall performance, offering the closest agreement with the analytical solution while maintaining stability and boundedness.

2.3.2 What do you think will happen to the solution at time approaches infinity with upwind and linear schemes and why?

With the upwind scheme, the solution will gradually diffuse over time due to strong numerical diffusion, eventually resulting in a nearly uniform scalar field. In contrast, the linear scheme introduces dispersive errors that cause oscillations, which may persist and grow over time, leading to a loss of the original Gaussian shape and reduced physical fidelity of the solution.

2.3.3 How did results change when you refine the mesh? Which scheme upwind or linear is more sensitive to the mesh refinement and why?

Mesh refinement reduces the numerical error for all schemes by decreasing the spatial discretization error. The improvement is more pronounced for the linear scheme than for the upwind scheme because the linear scheme is second-order accurate, whereas the upwind scheme is only first-order accurate. As a result, reducing the cell size leads to a faster reduction of error for the linear scheme, making it more sensitive to mesh refinement.

2.3.4 How did you choose a time step for simulations? What do you think is the effect of lowering or increasing it?

The time step was chosen based on the Courant number criterion to ensure numerical stability, with the Courant number kept below unity for all simulations. Lowering the time step reduces the temporal discretization error and improves the accuracy of the solution, while increasing the time step raises the Courant number, which can lead to stability issues and increased numerical error, potentially destabilizing the simulation.